

Brian Yue

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Technical Skills

Languages	Python, Java, C, SQL (PostgreSQL), NoSQL (MongoDB)
Infrastructure	Linux (Ubuntu Server), Docker, Git, Tailscale/WireGuard, Ollama
Concepts	REST APIs, Distributed Systems, Networking, Data Processing
Coursework	Advanced SQL, NoSQL, Operating Systems, Network Security, Artificial Intelligence, Computer Organization, Human-Computer Interaction

Technical Projects

Private AI Infrastructure & Network Security

2024 – Present

- Configured a self-hosted **Ubuntu Server** with **Tailscale** as a global **exit node**, securely routing 100% of traffic from all personal devices through a private encrypted tunnel.
- We deployed a network-wide DNS sinkhole using **AdGuard Home** to eliminate advertisements and tracking telemetry throughout the local and VPN-connected network.
- Implemented a monitoring stack with **Prometheus** and **Grafana** to visualize system health, network throughput, and container resource allocation in real-time.
- Local LLM inference was performed using **Ollama**, **LMStudio** and isolated from external APIs and cloud-based solutions to ensure total data privacy without data egress.
- Containerized **OpenWebUI**, monitoring tools, and network services with **Docker** to maintain a modular and scalable home-lab environment.
- Improved inference performance by **30–50%** using GPU passthrough and GGUF quantization, enabling efficient execution on constrained VRAM.

Multiplayer Educational Submarine Game — Team of 4

CS 2212

- Developed a Java-based side-scroller using **object-oriented design**, implementing custom physics and collision systems.
- Reached **90%+ unit test coverage** with **JUnit**, reducing regression issues across multiple releases.
- Led sprint planning and version control as **team lead**, managing task distribution, code reviews, and on-time delivery.

Selected Academic Projects

Low-Level System Architecture

Operating Systems

- Developed C-based utilities in a **Unix/Linux** environment, implementing thread management and process synchronization.
- Improved performance through concurrent execution and memory management.

Enterprise Mainframe Systems

IBM Z Xplore

- Completed systems administration and data management tasks in **z/OS**, including JCL scripting and dataset configuration.
- Achieved **100%** on final evaluation for enterprise computing labs.

Education

Western University — Bachelor of Science, Computer Science

Expected May 2026

Relevant Coursework: Operating Systems, Software Engineering, Object-Oriented Design & Analysis, Databases I, Computer Networks I

Experience & Leadership

Western Electronic Gaming Association (WEGA)

Sept 2024 – Present

Esports Executive

- Coordinated operations and event infrastructure for 10+ teams (~100 members), ensuring reliable execution of esports competitions.

Certifications

IBM Z Xplore: z/OS, JCL, mainframe systems